

Dataset Description:

The dataset selected for this analysis is sourced from the OGD **Open Government Data Platform**, specifically focusing on Socio-Demographic and Labor Statistics. The original dataset has been reorganized and shortened for this project specifically. The official metadata classifies this table (Table PCA) as "Basic Population and Socio-Economic Data." (in the official census website)

- **Number of Cases:** Each population category is under one of these cases—*age group, working-main and marginal, caste, literate, illiterate, non-working*
- **Number of Variables:** This dataset contains 49 variables.

COLUMN NAMES:

[1] "State code"	"Level"
[3] "Name"	"TRU"
[5] "No of Households"	"Total Population Person"
[7] "Total Population Male"	"Total Population Female"
[9] "Population in the age group 0-6 Person"	"Population in the age group 0-6 Male"
[11] "Population in the age group 0-6 Female"	"Scheduled Castes population Person"
[13] "Scheduled Tribes population Person"	"Literates Population Person"
[15] "Literates Population Male"	"Literates Population Female"
[17] "Illiterate Persons"	"Illiterate Male"
[19] "Illiterate Female"	"Total Worker Population Person"
[21] "Total Worker Population Male"	"Total Worker Population Female"
[23] "Main Working Population Person"	"Main Working Population Male"
[25] "Main Working Population Female"	"Main Cultivator Population Person"
[27] "Main Cultivator Population Male"	"Main Cultivator Population Female"
[29] "Main Agricultural Labourers Population Person"	"Main Agricultural Labourers Population Male"
[31] "Main Agricultural Labourers Population Female"	"Main Household Industries Population Person"
[33] "Main Household Industries Population Male"	"Main Household Industries Population Female"
[35] "Main Other Workers Population Person"	"Main Other Workers Population Male"
[37] "Main Other Workers Population Female"	"Marginal Worker Population Person"
[39] "Marginal Worker Population Male"	"Marginal Worker Population Female"
[41] "Marginal Worker Population 3_6 Person"	"Marginal worker Population 3_6 Male"
[43] "Marginal Worker Population 3_6 Female"	"Marginal Worker Population 0_3 Person"
[45] "Marginal Worker Population 0_3 Male"	"Marginal Worker Population 0_3_Female"
[47] "Non Working Population Person"	"Non Working Population Male"
[49] "Non Working Population Female"	

These include Population Categories—> *Population in the age group 0-6 Person, Literates Population Person, Total Worker Population Person, Main Working Population Person, Main Cultivator Population Person, Non-Working Population Person*; each of these are on the basis of STATE, TRU(total—rural—urban), and GENDER. And also has State, StateCode.

- **Total Population (Numerical):** Represents the total number of individuals in a region, further divided into male and female populations. This is the primary measure of population size.
- **Age Group (0–6) Population (Numerical):** Captures the number of children aged 0–6 years.
- **Literacy count (Numerical):** Includes literate and illiterate population counts

DATA CLEANING:

(left to add)

Analysis-1: Dependency Ratio (New Question add here)

Dependency Ratio:

a demographic indicator showing the ratio of dependents (those under 15 or over 64) compared to the working-age population (15–64). It signifies the economic burden on the productive workforce, with a lower ratio suggesting higher potential for economic growth.

Definition: It is calculated as $\left(\frac{\text{Children (0-14)} + \text{Elderly (65+)}}{\text{Working Age (15-64)}} \right) \times 100$.

Clearer Version: Measures the pressure on the working-age population to support the young (0–6) and the non-working population. Used by policymakers for social security planning.

OUR version of definition for analysis:

$\text{`Dependency Ratio`} = (\text{`Non Working Population Person`} / \text{`Total Worker Population Person'}) * 100$

Old Dependency Ratio (proxy: Total Pop – 0–6 – Workers)

$\text{`Other Dependents`} = ((\text{`Non Working Population Person`} - \text{`Population in the age group 0-6 Person'}) / \text{`Total Worker Population Person'}) * 100$

"Dependency Ratio" = Non-Workers per 100 Workers

This is the **standard economic measure** used for:

- Social security planning
- Labor force burden assessment
- Economic productivity analysis

Why this works:

So these are the 35 states we are working with:

[1] "JAMMU & KASHMIR"	"HIMACHAL PRADESH"	"PUNJAB"
[4] "CHANDIGARH"	"UTTARAKHAND"	"HARYANA"
[7] "NCT OF DELHI"	"RAJASTHAN"	"UTTAR PRADESH"
[10] "BIHAR"	"SIKKIM"	"ARUNACHAL PRADESH"
[13] "NAGALAND"	"MANIPUR"	"MIZORAM"
[16] "TRIPURA"	"MEGHALAYA"	"ASSAM"
[19] "WEST BENGAL"	"JHARKHAND"	"ODISHA"
[22] "CHHATTISGARH"	"MADHYA PRADESH"	"GUJARAT"
[25] "DAMAN & DIU"	"DADRA & NAGAR HAVELI"	"MAHARASHTRA"
[28] "ANDHRA PRADESH"	"KARNATAKA"	"GOA"
[31] "LAKSHADWEEP"	"KERALA"	"TAMIL NADU"
[34] "PUDUCHERRY"	"ANDAMAN & NICOBAR ISLANDS"	

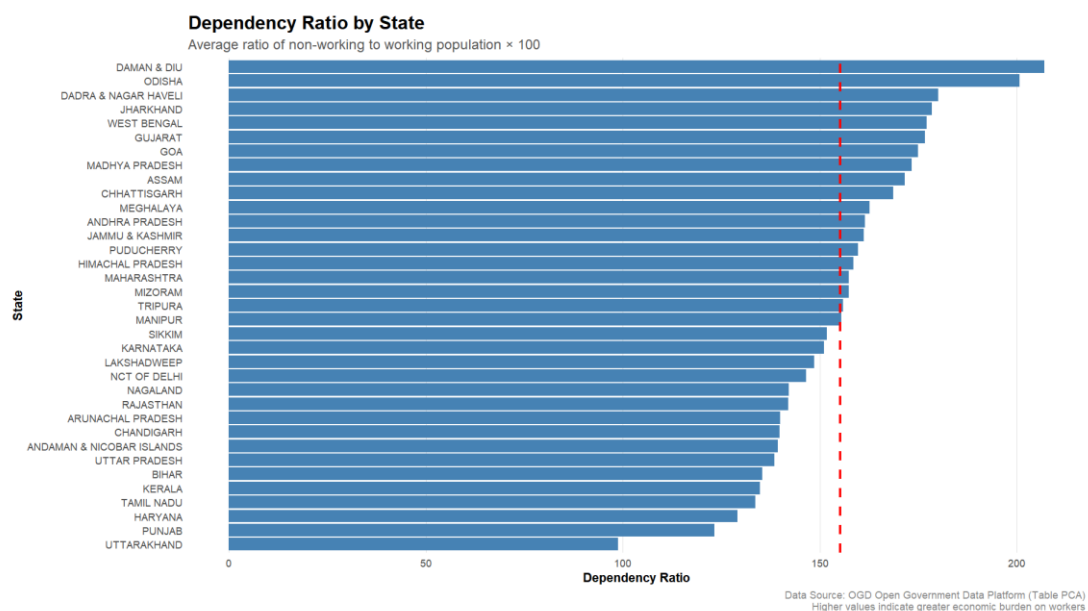
After adding the dependency ratios of different category:

	Name	TRU Total	Population Person	Population in the age group 0-6	Person
1	JAMMU & KASHMIR	Total	12541302		2018905
2	HIMACHAL PRADESH	Total	9108060		1593008
3	PUNJAB	Total	3433242		425897

4	CHANDIGARH Total	6864602	777898
5	UTTARAKHAND Total	6176050	712822
6	HARYANA Total	688552	65076
	Non Working Population	Person Total	worker Population Person Dependency Ratio Young Dependency Ratio
1		8218589	4322713 190.12572 46.70458
2		5994979	3113081 192.57382 51.17143
3		2223610	1209632 183.82533 35.20881
4		3305180	3559422 92.85721 21.85462
5		2886666	3289384 87.75704 21.67038
6		418514	270038 154.98337 24.09883
	other Dependents		
1		143.42113	
2		141.40239	
3		148.61652	
4		71.00260	
5		66.08666	
6		130.88454	

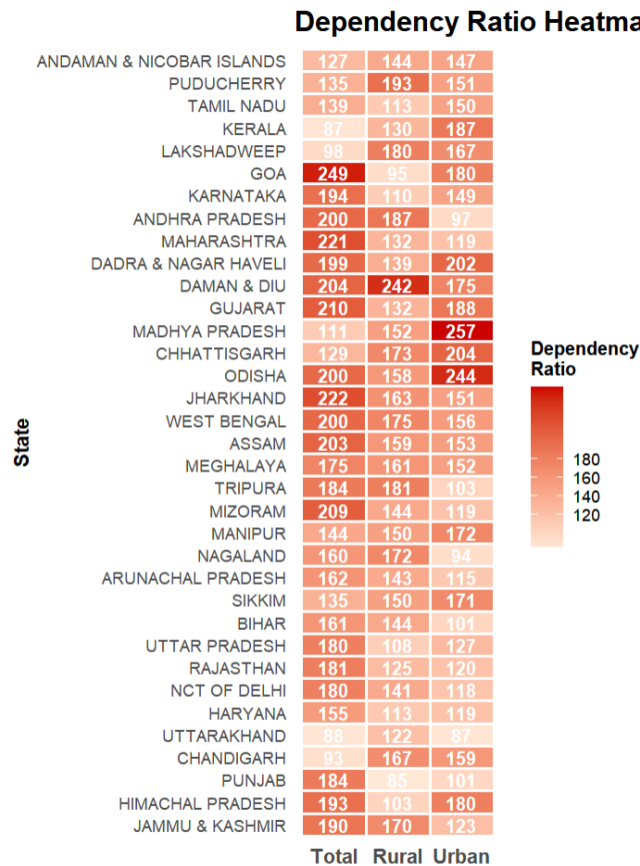
These are the summary stats:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
85.11	125.44	152.67	155.19	180.66	256.96



RED line indicates the national average dependency ratio = 155.1858

A heat map of the above data for each TRU(total rural and urban):



Summary stats for above TRU faceted heatmap:

```
# A tibble: 3 × 4
  TRU      `Mean Ratio` `Median Ratio`    SD
  <fct>      <dbl>      <dbl> <dbl>
1 Total      169.        180.  41.6
2 Rural      147.        144.  31.8
3 Urban      150.        151.  40.9
```

As you can see from the heat map:

TOP 5 STATES WITH HIGHEST ratio

Name	Average Ratio
1 DAMAN & DIU	207.
2 ODISHA	201.
3 DADRA & NAGAR HAVELI	180.
4 JHARKHAND	178.
5 WEST BENGAL	177.

TOP 5 STATES WITH LOWEST depend ratio:

Name	Average Ratio
1 UTTARAKHAND	98.7
2 PUNJAB	123.
3 HARYANA	129.
4 TAMIL NADU	134.
5 KERALA	135.

Conclusions from the top 5 and lowest 5:

Why high dependency?

- HIGH YOUNG POPULATION (0-6 age group):
 - Many children (need care, education)
 - Parents can't all work (childcare needed)
 - Example: Jharkhand has high fertility rate
- HIGH UNEMPLOYMENT/NON-WORKING:
 - Low education → fewer job skills
 - Agricultural seasonality → irregular work
 - Limited industries → fewer jobs

- Example: Odisha economy is agriculture-dependent
3. AGING ISSUES:
 - Elderly with no pensions
 - Need healthcare support
 4. MIGRATION:
 - Young workers leave for cities
 - Leaves behind children and elderly

LOW RATIO STATES (Uttarakhand, Punjab, Haryana):

why low dependency?

1. LOWER YOUNG POPULATION:
 - Fewer children per family
 - Better family planning
2. HIGHER EMPLOYMENT:
 - Better education → more jobs
 - More industries
3. MIGRATION INFLOW:
 - Young workers migrate IN
 - Increases worker pool
4. BETTER EDUCATION:
 - Higher literacy → better employment

We have also tried out the following Hypothesis test:

HYPOTHESIS TEST: Dependency Ratio vs Literacy Rate

Testing if high dependency ratio is linked to LOW literacy rate.

$r = 0.0595$ (Almost ZERO correlation!)

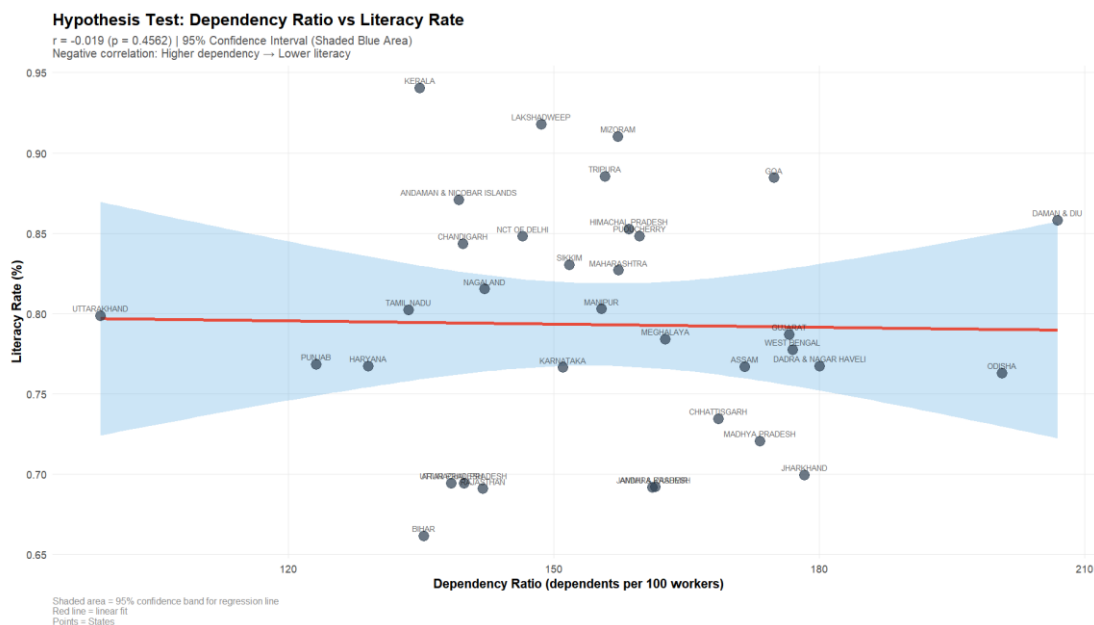
p-value = 0.6329 ($>> 0.05$, NOT significant)

CI = $[-1, 0.3367]$ (Includes ZERO)

SO therefore::

"Dependency ratio alone is NOT a strong predictor of literacy"

- This is valid research finding!
- Shows need for multi-factor analysis



Analysis-2: Workforce Participation

What is UPSS?

UPSS = Usual Principal + Subsidiary Status

This is an **official NSSO (National Sample Survey Office) classification** used in Indian labor statistics.

The 3 Categories in UPSS:

Status	Work Duration	Examples
Principal Status	6+ months per year	Regular jobs, continuous work
Subsidiary Status	30+ days in remaining months	Secondary part-time jobs
Marginal Status	< 3 months total	Occasional/seasonal work

NSSO uses this to measure **real employment** because:

- ✗ Census counts everyone who worked even 1 day = Misleading
- ✓ UPSS focuses on **meaningful employment** (6+ months or regular pattern)
- ✓ Separates stable workers from occasional workers
- ✓ Standard for policy comparisons across India

Our version of definition for analysis:

Using UPSS PROXY:

Your dataset doesn't have the exact NSSO categories, so you're creating a **proxy** (approximation):

UPSS_Proxy = Main Workers + Marginal 3-6 Workers

What You're Measuring:

"Workers with some regular employment pattern"

Component	Duration	Interpretation
Main Workers	6+ months	Definitely employed (stable)
Marginal 3-6	3-6 months	Semi-employed (half-year pattern)
UPSS Proxy Total	3+ months	"Regular enough" to count as employed

EXAMPLE to understand my code:

Main Workers (6+ months):	60 people
Marginal 3-6 Workers:	20 people
Marginal 0-3 Workers:	15 people
Non-Workers:	5 people

Total: 100 people

Metric	Value	Meaning
Principal Status Rate	60%	60% have stable year-round jobs
UPSS Proxy Rate	80%	80% have work for 3+ months/year
Marginal 0-3 Rate	15%	15% work less than 3 months
True Unemployment	5%	5% have zero work

Using Unemployment PROXY:

Official NSSO Unemployment Rate (UPSS):

Unemployment Rate = (Number of Unemployed Persons / Labour Force) $\times$ 100

- **Labour Force** = Employed (Principal Status + Subsidiary Status) + Unemployed
- **Unemployed** = Persons who were seeking/available for work but did **not** work even for 30 days in the year (subsidiary status criterion) + those unemployed on principal status.

What we used: $\text{Unemployment_Proxy} = (\text{'Non Working Population Person'} / \text{'Total Worker Population Person'}) \times 100$

Using Employment Quality Index:

What we are using: Positive = more main workers; Negative = more marginal workers 0-3

$\text{Employment_Quality_Index} = ((\text{'Main Working Population Person'} - \text{'Marginal Worker Population 0_3 Person'}) / \text{'Main Working Population Person'}) \times 100$

NSSO measures **underemployment** in two main ways:

- Proportion of usual-status workers who were **not employed** in the reference week (CWS underemployment).

- Person-days not utilised for work among workers (CDS).

Using LABOR DISTRESS INDEX:

$$\frac{(\text{Marginal Worker Population 0_3 Person} + \text{Marginal Worker Population 3_6 Person})}{\text{Main Workers}} \times 100$$

THEORETICAL BASIS:

Source 1: NSSO Employment Classification Framework

The National Sample Survey Office (NSSO) classifies workers by:

- Usual Principal Status (6+ months)
- Subsidiary Status (30+ days in remaining months)
- Marginal Status (< 3 months total)

This is the OFFICIAL Indian labor classification

Reference: NSSO 68th Round (2012), Ministry of Statistics & Programme Implementation

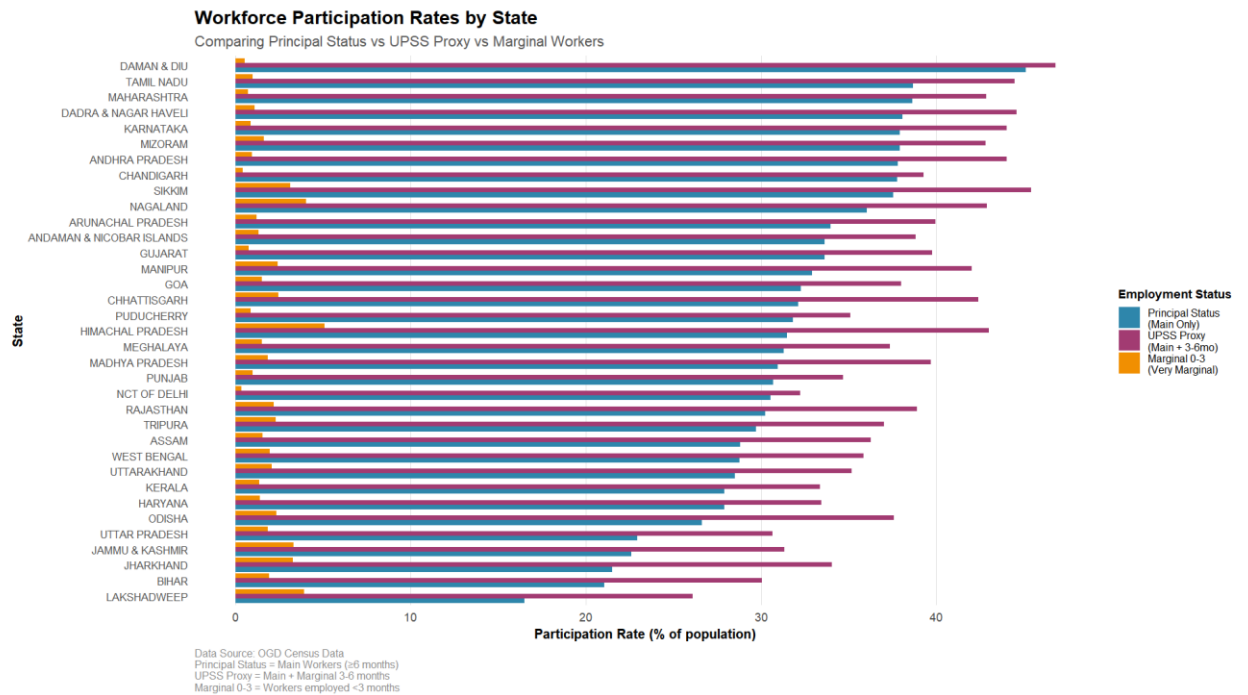
Source 2: ILO (International Labor Organization) Standards

ILO defines "precarious employment" as work with:

- Low wages
- Unstable/temporary nature
- No benefits/protections

Marginal workers 0-3+ 3-6 fit this definition perfectly

Below plot shows the participation rate of Principal Status, UPSS Proxy, and Marginal_0-3 workers per state



From this plot we can see that Difference in UPSS workers and Marginal_0-3 workers is very significant for every state,
Only Sikkim, Nagaland, Himachal Pradesh, Lakhsadweep have lower difference, meaning marginal workers are far more in comparison to other states.

The workforce participation rate BY TRU area:

TRU	Principal_Rate	UPSS_Rate	Marginal_0_3_Rate	Distress_Index	Quality_Index
<chr>	<db 1>	<db 1>	<db 1>	<db 1>	<db 1>
1 Rural	31.2	40.2	2.55	9.19	90.8
2 Total	31.5	38.9	2.09	7.43	92.6
3 Urban	31.9	35.8	0.86	3.08	96.9

CORRELATION between Labor Distress and Main workers, Marginal workers

Does Labor Distress correlate with Principal Status Rate?

Correlation: -0.7025

As distress increases, principal status DECREASES

Does Labor Distress correlate with Marginal 0-3 Rate?

Correlation: 0.9014

As distress increases, marginal workers INCREASE

Basically Low distress states have higher principal status workers

Labor Market Profile Heatmap by State

All metrics normalized (0-100) for comparison

State	Distress Index	Marginal 0-3 %	Main Worker %	Principal Status %	UPSS Proxy %
CHANDIGARH	22.5	1.5	88.5	32.5	22.5
NCT OF DELHI	15.7	1.5	93.7	30.0	22.5
DAMAN & DIU	5.5	1.5	95.5	45.1	45.5
MAHARASHTRA	7.5	0.7	89.5	38.7	42.5
GUJARAT	7.9	0.7	83.5	33.5	39.5
KARNATAKA	2.2	1.5	84.9	27.9	44.5
ANDHRA PRADESH	2.4	0.5	84.2	37.5	44.5
TAMIL NADU	2.5	1.1	85.4	38.7	44.5
PUDUCHERRY	2.5	0.9	85.5	33.8	35.5
DADRA & NAGAR HAVELI	2.5	1.1	83.5	35.1	44.5
PUNJAB	2.5	1.1	85.5	39.7	34.7
ARUNACHAL PRADESH	3.5	1.2	82.7	34.5	40.5
ANDAMAN & NICOBAR ISLANDS	4.5	1.5	83.9	31.5	27.5
MIZORAM	4.5	1.5	85.2	37.9	42.5
GOA	4.5	1.5	81.5	32.5	38.5
MEGHALAYA	4.5	1.5	85.5	31.5	37.4
KERALA	4.5	1.4	85.5	27.9	33.4
HARYANA	5.5	1.4	80.4	27.9	35.4
ASSAM	5.5	1.5	75.3	28.8	33.5
MADHYA PRADESH	5.5	1.5	75.5	30.5	37.2
RAJASTHAN	7.5	2.2	75.3	30.3	35.9
WEST BENGAL	7.5	2.2	75.2	28.8	35.9
UTTARAKHAND	7.5	2.5	77.5	28.5	35.2
MANIPUR	7.5	2.4	74.1	33.9	42.5
CHHATTISGARH	7.5	2.5	73.2	32.1	42.4
TRIPURA	7.5	2.3	75.5	25.7	37.5
UTTAR PRADESH	7.7	1.8	75.8	27.5	35.5
SIKIM	7.8	3.5	78.5	37.5	45.4
ODISHA	9.3	2.4	67.8	28.8	37.5
BIHAR	8.5	1.8	68.5	28.1	35.5
MIZORAM	10.7	4.5	72.5	35.1	42.5
JAMMU & KASHMIR	15.2	3.3	66.1	22.5	31.5
JHARKHAND	15.2	3.3	66.2	21.4	34.5
HIMACHAL PRADESH	22.7	4.5	67.5	31.5	43.5
LAKE SHADWEEP	22.7	3.5	67.5	31.5	43.5

Dark = High value, Light = Low value
Compare rows to see state patterns

- UPSS Proxy Rate (% of population): 34.03 %

ANALYSIS:

X CRISIS LEVEL - Extremely high employment precarity

✓ 60.2 % have stable jobs

△ 3.29 % in marginal employment

△ Below-average stable employment

HERE is the same for the best doing state in terms of wfpr emplyment and distress:

Rank 1 : CHANDIGARH

Labor Distress Index: 1.07

KEY METRICS:

- Main workers (% of total workers): 95.2 %
- Principal Status Rate (% of population): 37.77 %
- Marginal 0-3 Rate (% of population): 0.41 %
- UPSS Proxy Rate (% of population): 39.27 %

ANALYSIS:

✓ HEALTHY - Low employment precarity

✓ Strong majority (95.2 %) have stable jobs

✓ Very low marginal employment (< 0.41 %)

✓ Strong stable employment base

STRENGTHS:

- Stable labor market with secure employment
- Low unemployment and underemployment
- Strong formal sector presence
- Less vulnerable to economic shocks
- Better income security for workers

Rank 2 : DAMAN & DIU

Labor Distress Index: 1.25

KEY METRICS:

- Main workers (% of total workers): 95 %
- Principal Status Rate (% of population): 45.1 %
- Marginal 0-3 Rate (% of population): 0.54 %
- UPSS Proxy Rate (% of population): 46.8 %

ANALYSIS:

✓ HEALTHY - Low employment precarity

✓ Strong majority (95 %) have stable jobs

✓ Very low marginal employment (< 0.54 %)

✓ Strong stable employment base

STRENGTHS:

- Stable labor market with secure employment
- Low unemployment and underemployment
- Strong formal sector presence
- Less vulnerable to economic shocks
- Better income security for workers

WHat about Distress Index and Literacy Rate

=====

HYPOTHESIS TESTING: Distress Index vs Literacy Rate

Statistical Validation of Labor Distress Metric

'States with higher Distress Index should have lower literacy

NULL HYPOTHESIS (H0):

$\rho = 0$ (No correlation between Distress Index and Literacy Rate)

ALTERNATIVE HYPOTHESIS (H1):

$\rho < 0$ (Negative correlation: Higher distress → Lower literacy)

SIGNIFICANCE LEVEL (α): $\alpha = 0.05$ (95% confidence level)

EXPECTED OUTCOME:

Correlation coefficient < -0.6 (Strong negative)

P-value < 0.05 (Statistically significant)

FINDINGS:

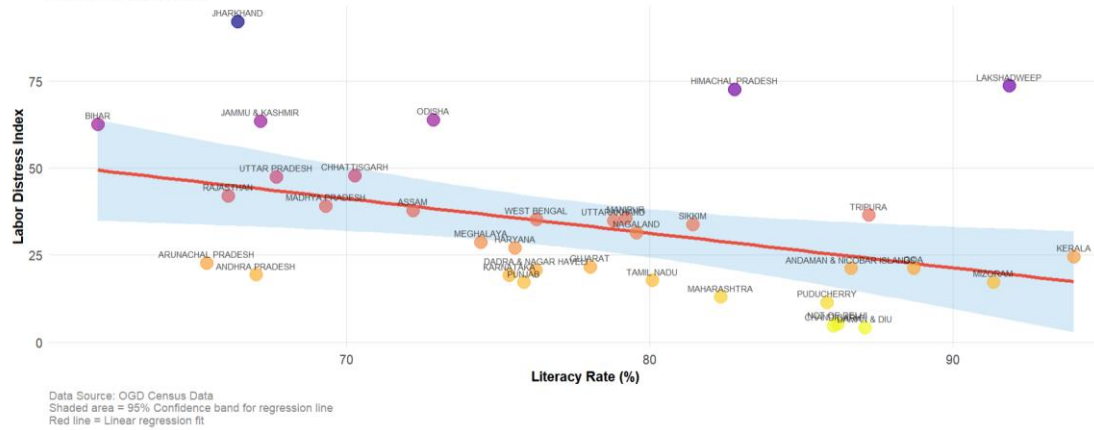
CORRELATION COEFFICIENT (r): $r = -0.4035$ ✓WEAK correlation, 16.28 % of variance explained

P-Value = 0.0081 ($p < 0.01$) – VERY SIGNIFICANT

Confidence interval: $[-1, -0.1362]$

Hypothesis Test: Distress Index vs Literacy Rate

Pearson $r = -0.404$ ($p = 0.0081$) | 95% Confidence Interval (Shaded)
PARTIALLY VALIDATED



CONCLUSION: ✓ HYPOTHESIS PARTIALLY VALIDATED

Relationship is statistically significant ($p < 0.05$)

But correlation strength is 0.4035037 not < -0.6

Still shows meaningful relationship, just weaker than expected

Employment Quality

TOP 5 STATES WITH BEST EMPLOYMENT QUALITY:

	Name	Quality_Index	Main_Workers_Pct	Distress_Index
	<chr>	<dbl>	<dbl>	<dbl>
1	CHANDIGARH	98.9	95.2	1.07
2	NCT OF DELHI	98.8	93.7	1.15
3	DAMAN & DIU	98.8	95.0	1.25
4	MAHARASHTRA	98.2	89.0	1.84
5	GUJARAT	97.8	83.6	2.19

These results definitely agree with the above results for Labour Distress, UPSS proxy workers and the Marginal workers analysis.

RESEARCH QUESTION-3

ANALYSIS-3: Agricultural Dependence per State

People in direct involvement in agriculture:

Agri_Workers_Person = `Main Cultivator Population Person` + `Main Agricultural Labourers Population Person`

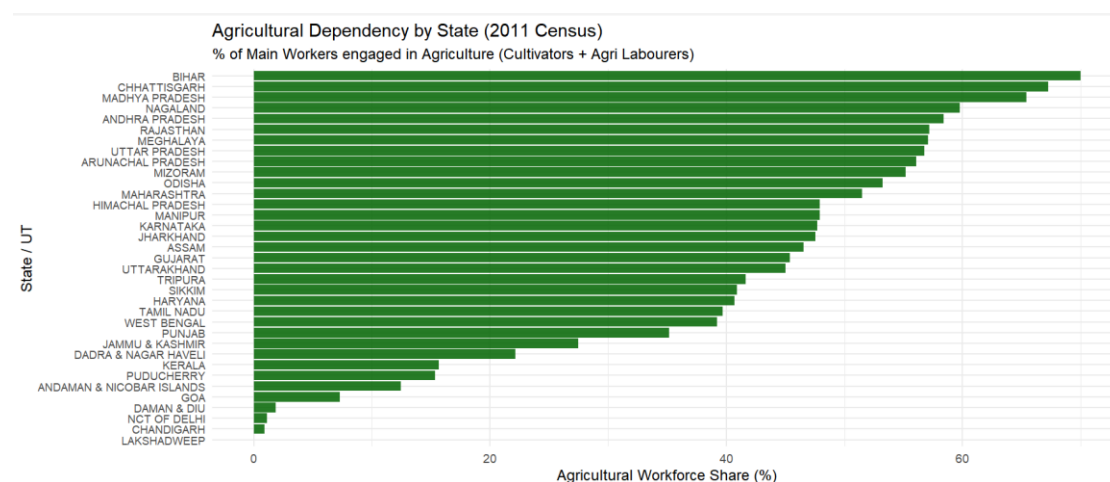
These are only main workers involved in agri.

Proportion of TOTAL POPULATION directly involved in agriculture:

Agri_Population_Share = (Agri_Workers_Person / `Total Population Person`) * 100

Agri_Workforce_Share gives the no of people involved in agri for 100 main workers:

Agri_Workforce_Share = (Agri_Workers_Person / Main_Workers_Person) * 100



This is a plot for agri workers relative to main workers, and on the basis of this

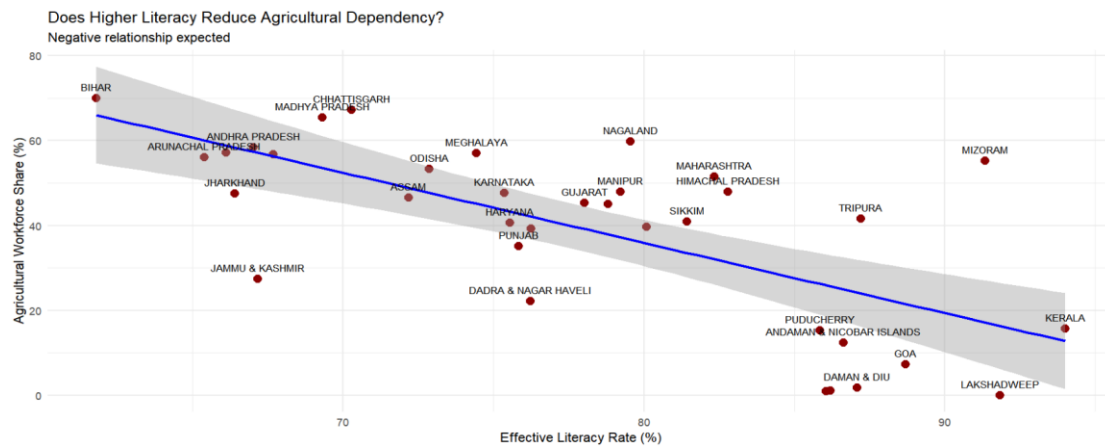
The bottom few States/UT which are involved in agriculture:

	Name	Agri_Population_Share	Agri_Workforce_Share	Literacy_Rate
31	GOA	2.3873106	7.314312	88.70210
32	DAMAN & DIU	0.8797642	1.837935	87.09879
33	NCT OF DELHI	0.3528306	1.116060	86.20878
34	CHANDIGARH	0.3377706	0.923745	86.04960
35	LAKSHADWEEP	0.0000000	0.000000	91.84697

Agriculture and literacy rate are closely related, since it's a common say that higher literate states get out of the direct agricultural work to go for further studies and higher education. So we tested this theory and it is positive.

Correlation coefficient (r) = -0.6846

The p-value = 0, which is clear from the scatter plot below, which shows a linear model and the corresponding correlation.



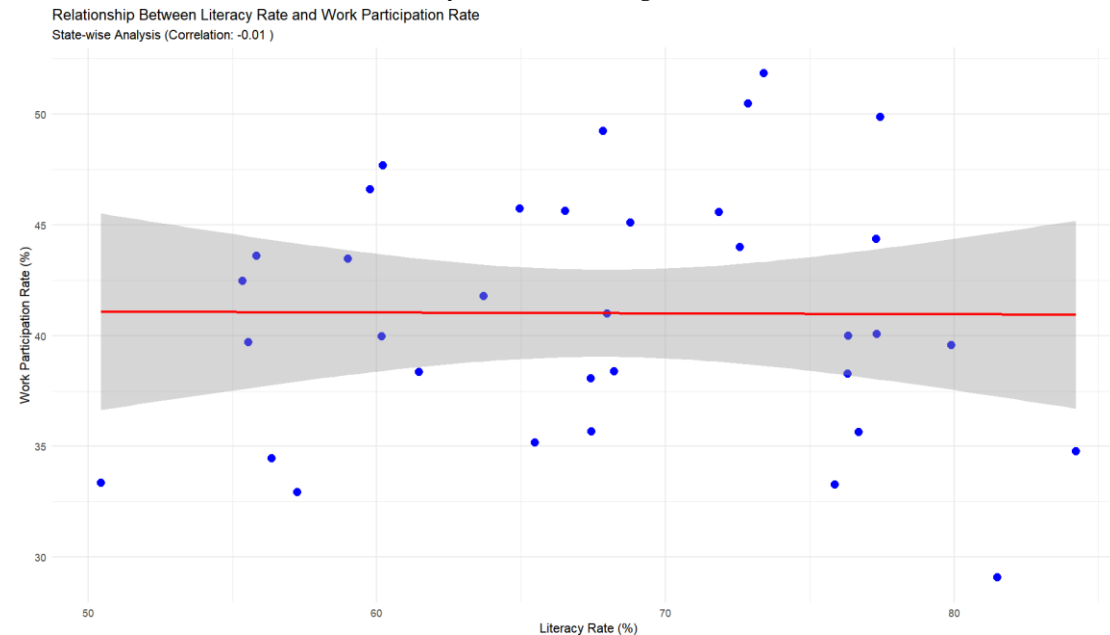
From this plot it is clear that Bihar has very high (highest) Agriculture Workforce Share and low Literacy Rate. More importantly we found out the Gender gap ($\text{Gender_Gap} = \text{Agri_Population_Share_Male} - \text{Agri_Population_Share_Female}$) is very high = 16.42455. Meaning male dominant agriculture share is prevalent here, with low literacy. Same is the case with Uttar Pradesh, a bit more Literacy Rate 67%, and high gender gap (16.57435).

Research Question-1

Analysis-4

Literacy rate versus Working Status: Are literate people more prone to get employed for periods of 6 months or more?

Correlation between Literacy and Working Status: -0.0062



The scatter plot of Workers and Literacy rate show low correlation almost straight line suggest that BUT if we look at main workers and marginal workers separately we see that:

Correlation with Main Workers: 0.304 means stable jobs are higher in areas with more literate people

Correlation with Marginal Workers: -0.4395 means unstable jobs in areas with illiterate people

Which are significant numbers, meaning these two cancel out to give the 0 correlation for the Overall worker category.

This proves that literacy in India doesn't have relation with quantity of workers, but it has relation with type of work.

So people in Higher literacy category are more prone to get employed in Stable Jobs (Main workers).